

Designing a Period Tracking App

PRO327 Final Written Assignment

Period 3 Student Led Project Report



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Abstract

Women's health has been historically marginalised in medical research, with the male body frequently being the default model. To this day, scientific research on the menstrual cycle remains limited. This project aimed to develop an application in which students can access scientific information about their menstrual cycle to better understand its influence on their daily life. The application was developed using the Flutter framework and coded in the Dart programming language. Figma was then used to create the app's design, integrating the researched information in an accessible way. The resulting product combines menstrual cycle tracking with day- and phase-specific content. This app includes details on energy and hormone levels, concentration, and nutritional suggestions, allowing the user to implement routines that align with their cycle. This project highlights that the use of digital tools can mitigate knowledge gaps by providing accessible scientific information for women and students.

1 Introduction

Throughout history, women's health has been underrepresented and taken less seriously in medical research. The male body has been frequently used as the default model, leading to significant gaps in the understanding of conditions specific to women, such as the menstrual cycle (Merone et al., 2022). For a long time, many taboos have surrounded the feminine reproductive system (Slawson, 2019). Consequently, research on the menstrual cycle remains limited, despite menstruation affecting half of the world's population. An Australian study found that less than 10% of women have a clear understanding of their menstrual cycle and how it affects their daily life (Medibank, 2023).

Due to the gaps in healthcare, research and technology, it is crucial to develop a tool for women to track their cycles and gain more education on their menstrual cycles. Currently, there are many period tracking apps available on the market, the most well-known ones being Clue and Flo. Both of these apps include a tracking page, and based on the data provided by the user, they predict the users' next period. Nonetheless, a limitation that both of these apps have is their paywalls, which significantly constrict the available information. Clue and Flo provide detailed information on menstrual health, fertility, and other crucial topics, yet to access it, a paid subscription is required. The free information is very limited and broad. Due to these paywalls, a clear gap is shown between menstrual health education and its

accessibility. For this reason, the final application will include all the information in detail and will be available free of charge.

To understand how and why energy, concentration, and mood vary throughout the menstrual cycle, it was necessary to examine the biology behind it. The menstrual cycle is a monthly process that prepares the body for the possibility of pregnancy. It typically ranges between 25 and 30 days, with an average of 28 days (Ferin, 2008). It is usually simplified into four main phases: menstruation, follicular, ovulation, and luteal (Cleveland Clinic, 2022b). Some phases naturally overlap; for example, the follicular phase runs from the beginning of menstruation until ovulation.

The hypothalamus-pituitary-ovarian axis is the regulating hormone system that controls the menstrual cycle by coordinating communication between the hypothalamus, the anterior pituitary gland, and the ovaries (Mikhael et al., 2019). The interaction between these organs causes physiological differences in each phase.

Menstruation: Throughout the cycle, many aspects are interlinked: hormone levels and the biological processes of each phase have consequences on how the body reacts. The menstrual cycle begins with menstruation, in which hormone levels are low, causing the shedding of the endometrium through the vagina. Energy levels will be low due to progesterone and estrogen levels being at their lowest (**Street2022**), and consequently, concentration will be negatively affected. It was demonstrated that when comparing men and women in the menstrual phase, there is a difference in the processing speed of information (Sawicka et al., 2025). The drop in progesterone and estrogen causes a decrease in serotonin, leading to low mood (The Belle Team, 2025).

Follicular: Through the follicular phase, the hypothalamus releases gonadotropin-releasing hormone (GnRH), which stimulates the pituitary gland to secrete luteinizing hormone (LH) and follicle-stimulating hormone (FSH) (Ferin, 2008). This causes multiple ovarian follicles to develop, one becomes dominant, while the other undergoes regression. The dominant follicle produces estrogen, which lowers the production of FSH and LH, then switches to a positive feedback loop, stimulating the pituitary gland to release LH and FSH. The peak in estrogen and FSH leads to increased energy levels (Fuller, n.d.), and working out becomes easier. In the early stages of the follicular phase, the brain is less networked and therefore foggy, whereas in the later stages of the follicular phase (specifically 2 days before ovulation) focus will improve (Aliva-Varela et al., 2024). The increase in estrogen also leads to an increase in serotonin and dopamine, resulting in better mood and a positive outlook(The Belle Team, 2025).

Ovulation: Ovulation begins 24 to 36 hours after the surge of LH (Ferlin, 2008). The follicle ruptures, and an oocyte is released into the fallopian tube. The ruptured follicle remains in the ovary and transitions into the corpus luteum. The surge in LH and FSH causes the ovaries to secrete estrogen and progesterone (Cleveland Clinic, 2022a), which results in a boost in energy (Fuller, n.d.). At this time, the brain is at its most dynamic state, leading to the highest concentration levels (Aliva-Varela et al., 2024) (Radyté, 2025). Attention span and memory perform the best, whilst processing speed shows no difference between women and men (Sawicka et al., 2025). The overall peak in energy and hormone levels will increase serotonin, causing a good mood, increased confidence, and positive outlook (The Belle Team, 2025).

Luteal: The luteal phase begins after ovulation and lasts an average of 14 days (Thiyagarajan et al., 2024). The corpus luteum produces high amounts of progesterone, while decreasing estrogen levels. This causes mood swings, anxiety, irritation, and anger for some individuals (The Belle Team, 2025). The uterus prepares for possible implantation by thickening the endometrium, but if no pregnancy occurs, the corpus luteum degenerates into the corpus albicans, a non-functional structure that causes the hormone level to drop. The endometrium is destabilized, leading to the next menstrual cycle (Hawkins & Matzuk, 2008). Lastly, throughout the luteal phase, the hormones fluctuate more, leading to a clear difference between the early and late stages of the phase. In the early luteal phase, energy levels still feel high as it is close to the ovulation period, and as progesterone peaks, the change of estrogen and progesterone levels will lead to suppression of gluconeogenesis. This makes it more difficult for the body to break down sugars (Fuller, n.d.), lowering energy level and exercise becomes more tiring in the late luteal phase. The brain is in a complex state; the pattern of brain activity is altered compared to ovulation, enforcing deeper thinking (Radyté, 2025).

As discussed, hormonal fluctuations throughout the menstrual cycle significantly impact daily life, affecting mood, energy, and concentration capacity. The goal for the app is to provide a platform where the information is easily accessible to each user, and bridge scientific knowledge with practical day-to-day experiences. In addition, it will provide a space where users can track their cycle and gain menstrual health education. The developed application focuses on students because of the proven impact that the menstrual cycle has on academic performance. The application will focus on the topics that are most influenced by this, such as energy, concentration, and mood. As well as how lifestyle choices can alter those, including information about food and vitamins that should be consumed. It aimed to provide users with a clear understanding of the cycle phases and support them with tips to anticipate and adapt accordingly.

2 Methods

It was decided to create a Flutter application, using Dart, which is an object-oriented programming language. These tools were chosen due to their ability to compile across platforms, specifically iOS and Android, as well as the extensive documentation available. The application can be designed directly in Dart, but Figma was chosen to allow for more freedom when designing the app. Using the FigmaToFlutter plugin, the elements designed in Figma could be converted directly into Dart code. In addition, GitHub was used as a version control and collaboration platform, enabling team members to work on different features simultaneously, track changes through commits, and resolve conflicts efficiently. This also provided a central repository for the project, ensuring that all code was backed up and that progress could be monitored throughout development (found in Appendix D).

2.1 Research phase

The team began by looking into research papers to understand what happens in each phase of the menstrual cycle. Google Scholar and other search engines were used to find relevant literature to understand the biology of the cycle. It was decided to separate the menstrual cycle into 5 phases: menstrual, follicular, ovulation, early luteal, and late luteal. Menstruation is part of the follicular phase, but it was separated for more clarity. Furthermore, the luteal phase was split into early and late due to the high fluctuation in hormone levels.

Based on the relevant information that was found throughout the research process, the topics that were chosen to be displayed in the application are expressed below:

- Hormone levels
- Effects of the hormones
- Energy levels
- Energy levels information
- What to eat
- Concentration
- Mood
- Types of vitamins

These fields were integrated into the application. They were specifically chosen to explain the scientific aspect of the menstrual cycle and provide targeted information, giving tips that can be applied to women's daily lives. As the application had a strong focus on students, it was necessary to highlight information that would be relevant and helpful for this target audience. For this reason, there was an emphasis placed on concentration levels, energy levels, and mood throughout the phases, as they strongly impact academic performance. For each of the fields, targeted information was gathered for each day of each phase and written in non-scientific language for easier comprehension by the users.

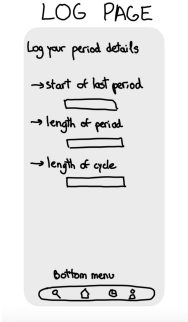
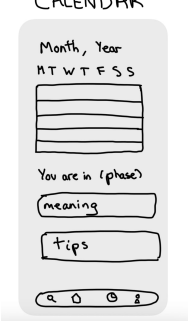
2.2 Initial design ideas and plan.

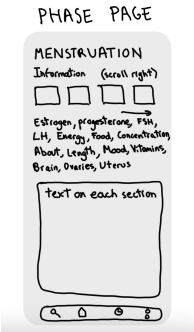
The goal of the layout of the application was to make it as user-friendly and intuitive as possible. To do this, attention was paid to making it easily scrollable, prioritising accessibility and attractive figures over complicated and condensed information.

After going through all the information found during the research phase, some general overviews of a possible design were discussed and drafted. Having multiple pages allowed for easy accessibility by the users as well as compartmentalising the information in an intuitive way. Creating several tabs gives access to deep insights without becoming overwhelming, allowing the user to easily find the information needed. The first drafts for each individual page can be found in Table 1.

Since the application was originally designed to work around the part of the cycle the user is in, it only showed information about that particular phase. It was decided to add additional pages, called Basic Information Pages, that would cover details regarding each phase, to allow users to learn about all parts of the cycle. This would ultimately be all the information the user would be looking for if they want to go dive deeper into insights about each phase.

Table 1: Initial design drafts showing each page of the application and a description of each one

Page Preview	Page Name	Description
	Log Page	A logging page where the user inputs the last period date, cycle length, and menstruation length. This data is used to personalise cycle predictions and daily recommendations.
	Home Page	The main page shown after logging in, displaying a circular diagram of phase progression within the menstrual cycle and providing brief insights into how the user may be feeling that day.
	Calendar Page	A visual calendar representation of the menstrual cycle, displaying predicted future cycles and highlighting important phase transitions.
	Daily Tips Page	Provides detailed daily information including cycle day, expected energy levels, hormone levels, health tips, nutrition recommendations, and insights into mood and concentration.

Page Preview	Page Name	Description
	Basic Information Pages	Contains overall educational information about each phase of the menstrual cycle, following the same structure and categories used throughout the application.

2.3 Algorithm

Since menstruation and cycle length can vary from woman to woman, and both affect the lengths of the different phases, these variables were taken as input from the user in the Log Page. During implementation, these variables were used, alongside knowledge about how these affect phase lengths, to create a *Calculate* class, which returns various relevant cycle parameters.

The *Calculate* class contains functions to compute the date of the next period, taking into account the last period logged and the length of the cycle input. The day of the cycle is also calculated by finding the difference between the date of the last period and today's date. A third function returns the phase and day of the phase based on the day of the cycle.

The most logical storyline was to develop the application in a way that it adapted dynamically to the phase of the cycle the user is in, contrary to storing static information. This was done by integrating .json files into the Dart code, which contained all the relevant information.

A file for each phase was first created in Excel; each row corresponded to the day of the phase, which started at 1 and continued until the highest day within a healthy range. Each column had specialised information and tips depending on the day of the phase. This was done to allow for fluctuations in the phases in all individuals.

For the menstruation phase, the maximum is 8 days (Reed & Carr, 2018). The follicular phase can be a maximum of 21 days (Cleveland Clinic, 2022a). It is crucial to note that when creating the Excel sheets, the menstrual and follicular phases have been split up, even though they biologically overlap. This resulted in a maximum 14-day healthy length for the follicular phase. The rest of the phases, ovulation and

luteal, do not vary, remaining at 1 day and 14 days respectively (Cleveland Clinic, 2022c). Due to the separation of the Excel sheets, the early luteal phase was kept to a maximum of 6 days, whilst the late luteal phase was 8 days (Sawicka et al., 2025).

The Excel files were converted into .json files to allow for easy integration and parsing of this nested data based on the current day and phase. The .json files were uploaded into the *assets* folder, and declared into the *pubspec.yaml* file.

The .json files were accessed within a specialised class, which finds the corresponding .json file of the phase and returns the information based on the day of that phase. The logic followed in order to display the information to the user in a dynamic way is found in Figure 1.

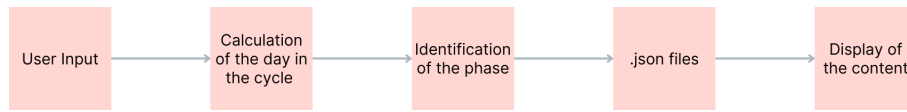


Figure 1: Class diagram of the overall architecture of the application.

In addition to the .json files for each phase, a general “All Phases” .json file was created using the same process and containing the same fields. This file was used to display more general information in the Basic Information Pages, which only contain broad details about the phase, not specialised by day.

2.4 Figma and designing the app

Figma was used to convert this idea and sketches (Table 1) into a programmable design. This tool supported the collaborative nature of the project, providing a shared visual framework that the whole team could discuss and make use of. The development of the interface design was based directly on research done and fields chosen. Using the original sketches shown in Table 1, the Figma designs were developed incorporating figures and icons created by (Icons8, 2024). The priority when designing was making the text clear, readable, and informative without being overwhelming.

These became the blueprint for the coding team. The text containers were linked to the .json files, therefore avoiding loss or misleading information. Figma allowed for the structuring of textboxes in a visual way and bridged the scientific content with its technical implementation.

The key design feature when using Figma was its scrollable sections, both vertical and horizontal, so that the large volume of information could be accessible without becoming overwhelming. Content that was horizontally scrollable was referred to as

"scrollable widgets". This tool was implemented for the fields, such as hormones, energy levels, nutrition, and mood. Scrolling became useful because it segmented the information into units, making it readable.

Components in Figma were left consistent throughout the pages so that the application looked cohesive, ultimately becoming the intermediate that links design with the app's interface and development. It was also ensured that the navigation between pages would be easy and intuitive, adding a Navigation Bar and buttons connecting the pages.

2.5 Testing

The final step of this process was to adapt a test-driven approach to refine the app's functionality and design, and potentially identify hidden faults. The team adapted a manual process with three stages: writing detailed test-cases according to Figma designs, implementing features, manually executing the tests to compare actual and expected outputs, and refining the code handle failures and improve reliability. The testing ensured that edge cases, such as empty input, max length, and special characters, were properly handled. Additionally, input validation and error handling were thoroughly tested to confirm that incorrect or out-of-range data were consistently rejected by the application. Finally, all test cases were grouped under pages of the app, and were organised in tables that followed the structure of Table2 (the full table can be found in the Github repository in Appendix D). Each test case was feature-specific and categorised in either functionality or user-interface.

Table 2: Test case table headers

Page	Test Case ID	Test Description	Input	Expected Output	Actual Output	UI / Functionality	Status
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3 Results and Discussion

3.1 Implementation

Dart is an object-oriented programming language, the code of the app was split into classes that fall under 3 categories: algorithms, classes for each page in the app, and helper classes, which include designs for buttons or text boxes. All classes can be found in Appendix A.

An overview of the architecture of the application is found in the class diagram below (Figure f2). This diagram shows the main classes used in the code. The user interacts with the four main classes under the Primary User Interface category, which correspond to each page of the app. The users can navigate between these pages using the *MainNavigationBar* class (which has the *NavigationState*). After logging their details in the Log Page, the *Calculate* class computes the relevant cycle parameters based on the input from the user. These parameters are used in the *CycleDataProvider* class, which accesses the .json file corresponding to the correct day of the phase that the user is in (Figure 3). The information returned from this class is displayed in the Daily Tips and Home Pages.

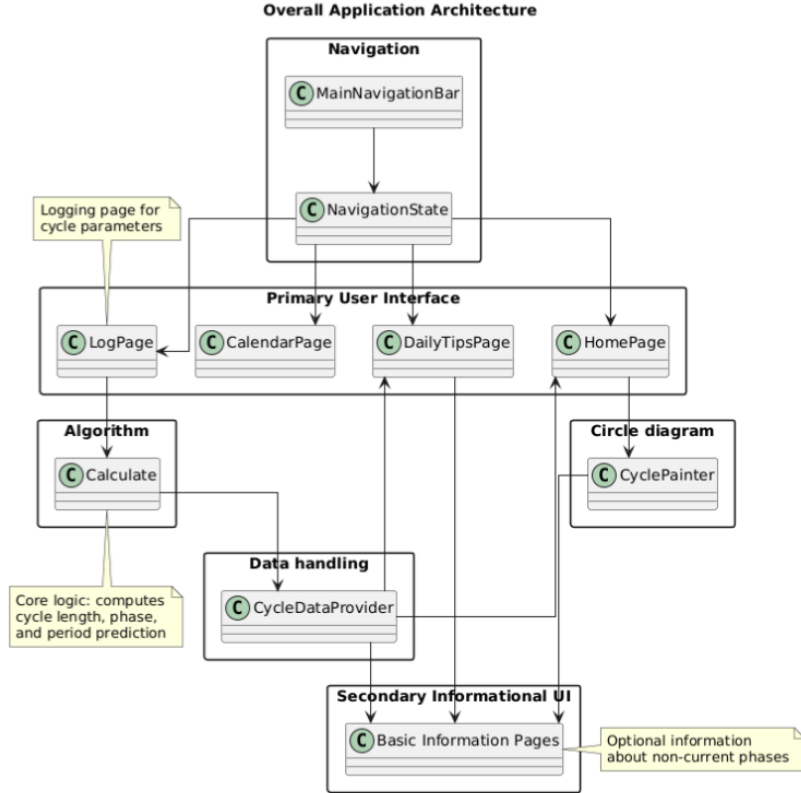


Figure 2: Class diagram of the overall architecture of the application.

In order to access information about non-current phases, the user can open the general Basic Information Pages using the circle diagram in the Home Page or the horizontal scroll element present at the bottom of the Daily Tips Page.

The *LogPage* class extends a *StatelessWidget* and calls the *LogCalendar* class, which is a *StatefulWidget* class, meaning it changes depending on how the user interacts with it. The main purpose of this page is to allow the user to log cycle-related information, such as the last period date, the length of their period, and their cycle. Upon submission, data is stored using the *SharedPreferences* (Flutter Documentation, n.d.) plugin and immediately reflected over other parts of the application, including the Home Page, Daily Tips Page, and Calendar Page.

During testing, all recorded data remained persistent after restarting the application, indicating that the logging functionality operates reliably and supports continuous long-term tracking. However, testing also revealed that the input validation is limited: the application currently allows users to select dates from years in the past or enter unusually large values for period or cycle length, which are not realistic and could lead to inaccurate calculations.

The second class is the *HomePage*, which extends a *StatelessWidget*. It acts as the main dashboard of the app and provides quick access to key features. This page includes a circular diagram with an updating cursor showing the user's current position within their menstrual cycle, which changes dynamically as new period data is logged. The circular shape is divided into 5 sections, representing the phases of the cycle. It is rendered using a *CustomPainter* class, which draws the circle and its segments. Each section functions as a button, allowing users to tap it and navigate to the corresponding Basic Information Page for more detailed information about that phase. Within this representation, the day of the cycle, current phase and calendaristic date are displayed. Additionally to this visual, the Home Page also includes information about the expected future period date, the day of the phase and a brief summary of the day.

Test cases confirmed that the cycle calculations worked properly when valid inputs were entered in the Log Page. The circular diagram adapts dynamically with different cycles and period length, and an indicator moves along it across different cycles, including edge cases like month boundaries and leap years. Interaction between pages was also tested, making sure that tapping on one of the sections of the circle would lead the user to the correct information page. Some inconsistencies were found when tapping on the boundary between phases.

The third page is defined in the *DailyTipsPage* class, which also extends a *Stateless-*

Widget. It provides users with daily phase-related information and recommendations based on their position in their cycle. The display content includes information specific to the day within the current phase and tailored guidance related to energy and well-being, along with insights into the anatomy of the body. Furthermore, graphs are included to show how the relative hormone and energy levels change throughout the cycle. This information is determined using the cycle data calculated from previously logged period entries. From this page, non-current phase information is accessible, within the Basic Information Pages. During testing, changes in logged data resulted in corresponding updates to the displayed phase and recommendations, demonstrating that the page remains synchronized with the underlying cycle calculations and provides relevant daily tips to the user.

Finally, the *CalendarPage* is the only one that was implemented as a *StatefulWidget*. This is because it needs to update dynamically as the user interacts with it. It provides a monthly overview of predicted cycle dates, allowing users to easily identify patterns over time. Predicted menstruation days are visually highlighted, while the uncertainty days before and after are marked using a different indicator. The healthy range which a period can vary is ± 5.3 days for people aged 20 (Harvard T.H. Chan School of Public Health, 2022). Alongside menstruation, ovulation days are also signalled in the calendar. Moreover, users can navigate between months and select individual dates to gain information about the predicted phase, including summary information about it.

During testing, newly logged period dates were immediately reflected in the calendar view, and predicted dates adjusted accordingly, indicating that the calendar remains synchronized with stored data. This confirms that the feature functions as intended and supports effective cycle tracking.

The overall testing process was largely successful. Out of 85 planned test cases, the final version of the application passed 78, demonstrating strong functionality and adherence to the intended design features. Most of the failed tests were due to the application not rejecting invalid input, such as zero or excessively large numbers entered on the *LogPage*, highlighting an area for improvement in input validation.

3.2 Algorithm

The *Calculate* class contains all the functions which compute the relevant parameters based on the user input. This class is of the type *ChangeNotifier*, allowing its variables to be accessed by other classes throughout the application. The first function within the class calculates the next period based on the user's input of

their last period date. This is simply done by adding the cycle length to the date of their last period, and returning the output calculated. The second function takes the date of the user’s last period as an input and finds the difference in days until today’s date, obtaining the day of the cycle. The *determinePhase* function takes the day of the cycle, the length of the cycle, and the period length as input, and uses these to return the current phase and the day of the phase; an overview of this function can be found in Table 3. This is done by first setting the ovulation day as the 14 days before the end of the cycle, since this is the average (Cleveland Clinic, 2022b). The phases and days of the phase are calculated as shown in Table 3.

Table 3: Overview of the *determinePhase* function within the *Calculate* class

Phase	Condition for the day of the cycle	Day of the phase
Menstruation	$\leq periodLength$	Day of the cycle
Follicular	$< ovulationDay$	$cycleDay - periodLength$
Ovulation	$= ovulationDay$	1
Early Luteal	$\leq ovulationDay + 6$	$cycleDay - ovulationDay$
Late Luteal	else	$cycleDay - ovulationDay - 6$

3.3 Retrieving Data

To be able to make the text dynamically change depending on the phase, day of phase, and the field that the user wishes to see information about, a *CycleDataProvider* class was employed, which manages phase-specific information from .json files. This class works together with a *PhaseDayInfo* and the *Calculate* class in order to obtain the biologically accurate information from the files (as shown in Figure 3). The *PhaseDayInfo* class represents information in all fields from a single day of the phase. It contains the phase, respective day, and all of the information from that day.

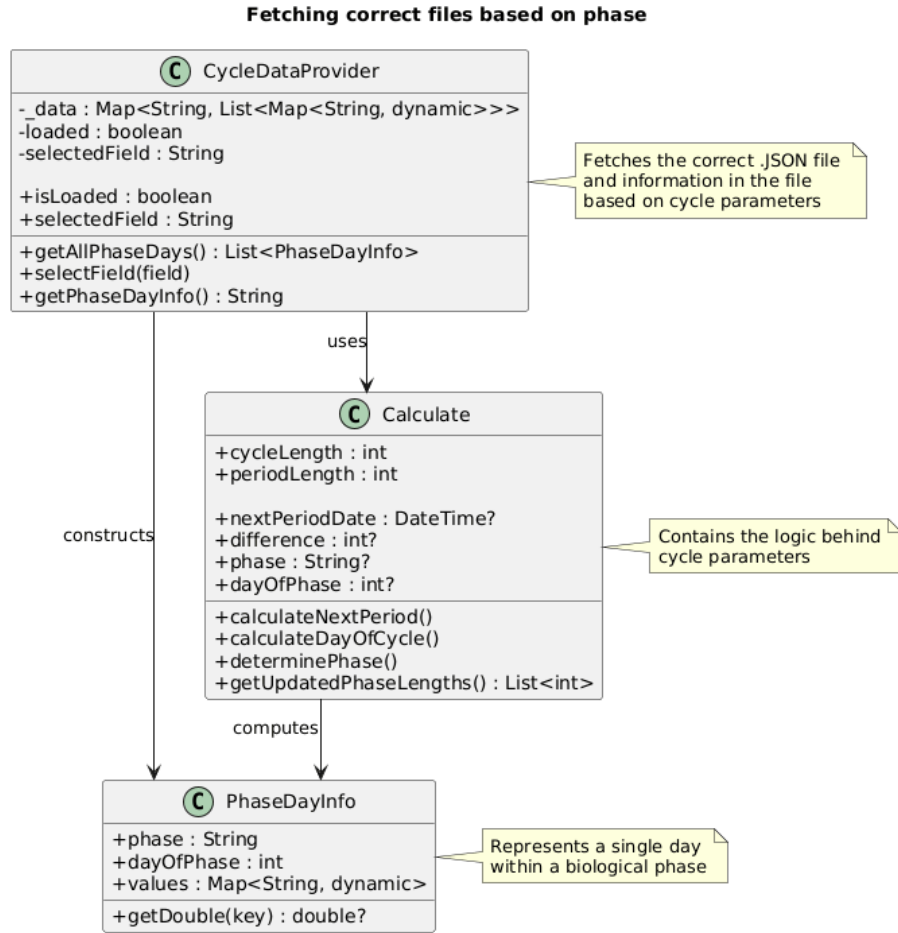


Figure 3: Class diagram describing the classes involved in parsing and fetching the information from the .JSON files.

CycleDataProvider decodes and stores all of the .json data in memory as the `_data` variable. For each phase, it creates a *PhaseDayInfo* object for each day of the phase, and returns a list of *PhaseDayInfos*. *getPhaseInfo()* method selects information from the selected field for a single day, for user interface display.

The *Calculate* class computes the relevant cycle parameters based on the users' input (Table 3). The current phase and day of phase computed in this class is used to fetch the correct .json entry.

3.4 Information displayed

The content of the .json files were similar to the ones that were previously mentioned in the introduction. The main differences were that additional fields were added, such as food and recipes, today's recap, and the insights into anatomy. Additionally some of the fields were separated to allow the information to be more detailed. For

instance, the field including hormone levels was separated into a total of 4 columns, one for each hormone expressed in the cycle. It was decided to use a numerical scale for expressing variable data, specifically hormone and energy levels. Initially, the data points were expressed by stating whether the levels were “low, high, medium, decreasing or increasing” yet assigning a numerical scale allowed for the data points to be graphed, therefore serving as a visual explanation of how these factors vary throughout the days of the cycle. The numerical scale was determined as 1, the lowest level; 5, the peak; 4, high; 3, medium; 2 when it was decreasing or increasing.

To plot these values, a for loop iterates through all of the phase-specific .json files in order (Menstrual, Follicular, Ovulation, Early Luteal, Late Luteal) and appends the day and its respective hormonal level to a nested list of double-type variables. The resulting list contains data points, where the first value of the double is the x value (day of cycle) and the second is the y value (hormone level). These values are plotted as a line graph in Dart, where all of the hormone levels appear on the same graph with different colors to allow for a direct comparison.

The “Hormone Effects” field was separated into 4 separated categories, each one containing detailed information about the impacts of estrogen, progesterone, LH, and FSH. Each of the columns included an explanation of the role and effects the hormones have throughout the cycle. The anatomy aspect was also split into 3 different fields to provide detailed physiological information about the body parts most affected by the menstrual cycle: the brain, ovaries, and uterus. An explanation of the biological background of each of those body parts was added.

Furthermore, additional categories were introduced, and the final fields, alongside a brief explanation of the information they covered is found in Table 4. To make the application more straightforward, a summary column providing a simplified explanation of the phase and a daily overview describing what the user may feel throughout the day were added. Finally, the fields were renamed and simplified to be used as labels of the buttons of the horizontal scrollable widgets within the application.

Throughout the research stage, inconsistencies between sources highlighted the difficulties of finding accurate and reliable information on the menstrual cycle. Several sources expressed that the luteal phase had a fixed duration of 14 days, including physiology papers (Thiyagarajan et al., 2024) or endocrinology articles (Reed & Carr, 2018). The project had started with this reference length and was kept through the duration of the project.

These inconsistencies reflect the limitation in menstrual cycle research and sources

Table 4: Each field that was included in the Excel sheets

Fields	What is included in the fields
Hormone levels	Levels of estrogen, progesterone, FSH, and LH levels
Effects of the hormones	Explanation of what estrogen, progesterone, FSH, and LH do
Energy Levels	A scale of how energized you may feel
Energy Levels Information	Explanation of how energized the individual might feel in each phase
What to eat	Information on types of foods to eat, with some examples
Food and Recipes	Presents a link to a website that has more specific foods to eat, as well as recipes for each phase
Concentration	Explanation of the concentration levels per phase
Health aspects	Information on changes that can occur in the cycle and what these could mean for your health
Mood	Insight into the possible moods per phase
Types of vitamins	Recommendations of vitamins to take during each phase
Insights on anatomy	Information about what happens in the brain, ovaries, and uterus
Phase information	Explanation of what occurs in the actual phase
Today's recap	A brief summary including a sentence of what happens regarding energy, concentration, and mood

and the lack of documentation. This lack of accurate information and inconsistencies between sources posed a challenge for the research behind the application.

3.5 Final arrangement of app

The app's design aimed to provide quick and efficient access to information. This is demonstrated in Figure 4, which highlights the design approach in which each page is at most two clicks away from the user at all times.

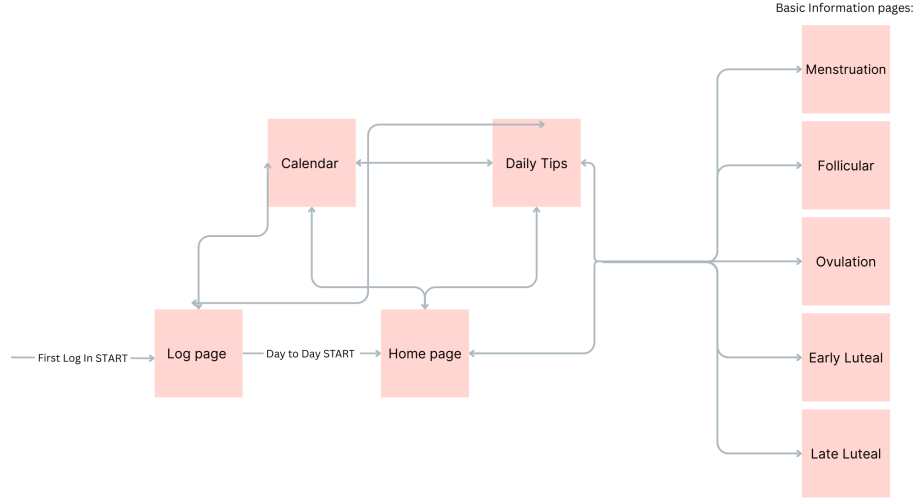


Figure 4: The Apps' flow from page to page.

Figure 4 displays the app's navigational structure and the relationship between pages. When signing into the app for the first time, it will open on the Log Page. After submitting the details, the user is automatically directed to the Home Page. After the initial sign-in and detail submission, all subsequent app launches will open directly on the Home Page.

The Home Page functions as the main center for the user. From this page, the user can access the Calendar and Daily Tips pages through the app's menu, which allows for seamless navigation between Home, Calendar, Log Page and the Daily Tips Page. Additionally, the Basic Information Pages can be accessed when clicking on their respective phase on the Home page circle as well as from within the Daily Tips page, as illustrated by the arrows in Figure 4.

3.6 Final Design

Screenshots of each page of the app is found in Appendix C. The final Figma designs are found in Appendix B. All icons and figures of women presented were created by Icons8 (Icons8, 2024).

4 Future Developments

As more research was conducted, it was observed that the duration of the luteal phase may vary, often with an average between 12 and 14 days (Cleveland Clinic, 2022c). However, due to discrepancies within the data gathered, and the coding backbone already being completed, the fixed length of 14 days was kept. The variability of the luteal phase highlights a key area in which the app is currently limited. As further research on the menstrual cycle is conducted, the aim is to have the app adapt to incorporate the new findings into the interface. Future developments of the app should include enhanced data storage and analytical capabilities, providing the necessary framework for a model that predicts future periods based on the user's previous logs. This allows the past cycle days to be displayed directly on the calendar page. Additionally, users would also be able to log energy levels, mood and workouts, through which the app would adapt and personalise its tips and information displayed. This data could be used to train a Large Language Model, which would be able to predict upcoming symptoms based on the phase, helping users plan and prepare for exams or other academic commitments.

A further improvement would be to create a Personal Page within the app. This should take in inputs such as age, medication, and health conditions, adjusting the information within the app accordingly. Additionally, it would be possible to track the user's symptoms such as cramps and headaches. This information would allow it to predict how the user will feel in the upcoming cycle. Another improvement of the app would be to create a page where scientific articles are linked, allowing users to have access to more in depth information. Finally, a platform could be added within the app in which users can communicate with each other and securely talk to a healthcare professional.

5 Conclusion

A variety of period tracking apps have been available for years, however, many are limited by paywalls and lack of phase specific information. This project aimed to design and program an evidence-based period tracking app that supports users throughout the different phases of the menstrual cycle. The app was successfully built with all the main features originally planned. It provides users with information about energy levels, mood, and concentration throughout the different phases of the menstrual cycle, while also educating them on the underlying physiological reasons by displaying hormone fluctuation and energy levels in graphs.

Users input their last period start date alongside other information. This allows the app to calculate the user's current period phase and predict future period dates. The planned design was successfully integrated into the code, ensuring a consistent and user-friendly experience. Overall, the app meets its main goal of educating users about their menstrual cycle.

The main limitation of the app is the lack of data storage functionality. Currently, users can only input a single period date, which does not allow for tracking of previous cycles. Additionally, there are some discrepancies in information obtained from different sources, leading to some issues when calculating some parameters like the length of the luteal phase.

Further development could focus on widening the tracking and prediction capabilities, adding the tracking of different parameters like mood or energy levels, and using these for more advanced cycle predictions.

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6 Appendix

A Overview of classes

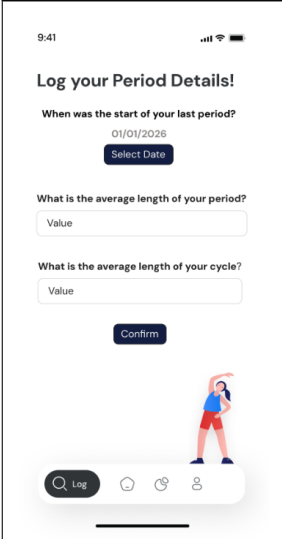
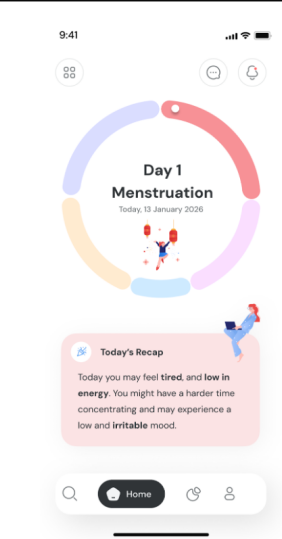
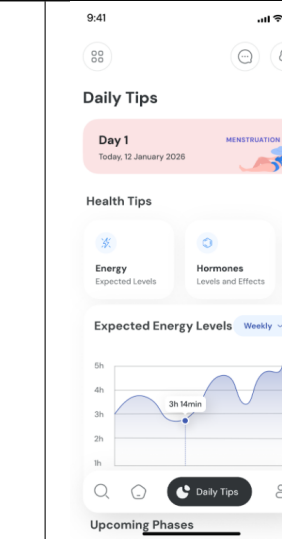
Table 5: Overview of application classes and their inheritance

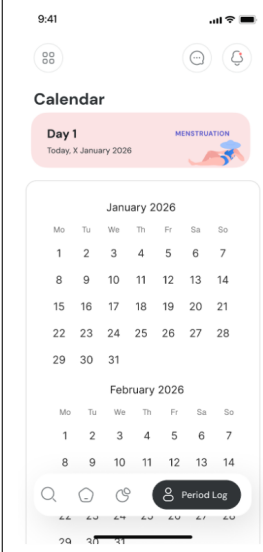
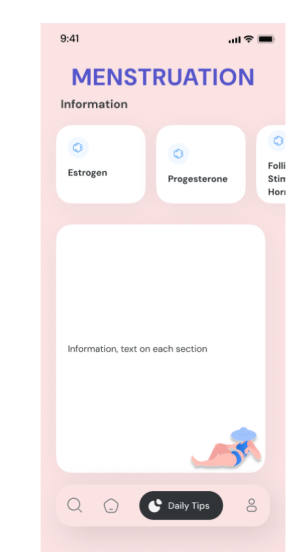
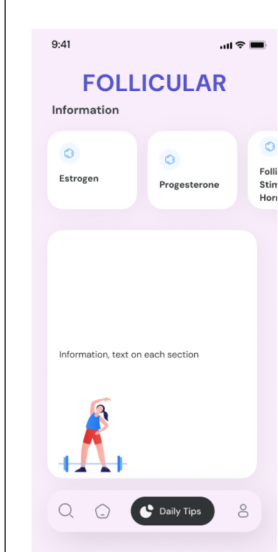
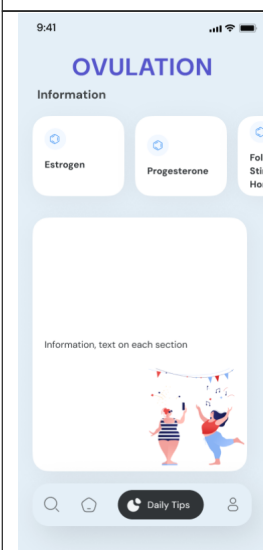
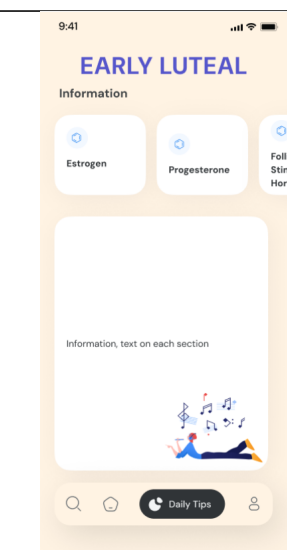
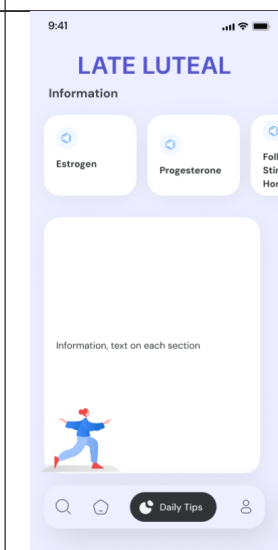
Class name	Extends
MyApp	StatelessWidget
PhaseDayInfo	n/a
CycleDataProvider	ChangeNotifier
Calculate	ChangeNotifier
MainNavigationBar	StatefulWidget
CustomNavigationBar	StatelessWidget
_NavButton	StatelessWidget
_MainNavigationBarState	State<MainNavigationBar>
LogPage	StatelessWidget
_LogCalendarState	State<LogCalendar>
HomePage	StatelessWidget
CyclePainter	CustomPainter
DailyTipsPage	StatelessWidget
CalendarPage	StatefulWidget
_CalendarPageState	State<CalendarPage>
MenstruationPage	StatelessWidget
FollicularPage	StatelessWidget
OvulationPage	StatelessWidget
EarlyLutealPage	StatelessWidget
LateLutealPage	StatelessWidget
SelectButton	StatelessWidget
TextBox	StatelessWidget

Class name	Extends
HorizontalScrollButton	StatelessWidget
HorizontalScrollButtonPhases	StatelessWidget
_LegendItem	StatelessWidget

B Final Figma design

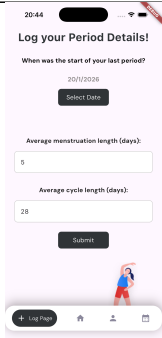
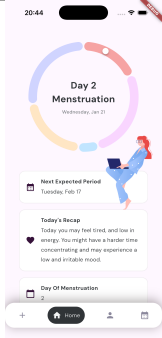
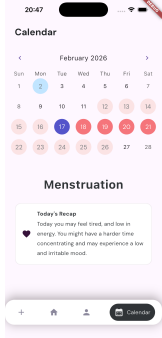
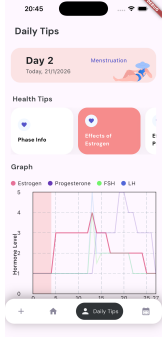
Table 6: User interface pages of the application

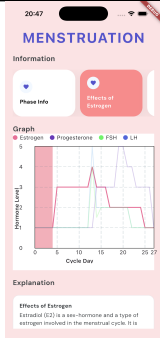
Log Page	Home Page	Daily Tips Page
		
Calendar Page	Menstruation Page	Follicular Page

Log Page	Home Page	Daily Tips Page
		
Ovulation Page	Early Luteal Page	Late Luteal Page
		

C Final Application design

Table 7: Initial design drafts showing each page of the application

Page Preview	Page Name
	Log Page
	Home Page
	Calendar Page
	Daily Tips Page

Page Preview	Page Name
	Basic Information Pages

D Link to GitHub repository

<https://github.com/ioanasarah/Design-Period-Tracking-App.git>